Developing an Automated GaN Quantum Dot Detection Pipeline

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Why automate quantum-dot detection?

- GaN quantum dots appear as nanoscale features in AFM scans.
- We want to extract quantities such as
  - dot position,
  - height,
  - lateral size,
  - dot density.
- Manual labelling is slow and subjective, especially across many scans.

Goal: build a reproducible pipeline that turns an AFM height map into detected and measured QDs.

Insert example AFM scan here

Raw scan + labelled QDs

The pipeline

Raw AFM scan → Preprocessing → QD detection → Height filter → Evaluation

Preprocessing

- remove scan tilt / background variation,
- reduce line artefacts and noise,
- preserve QD peaks.

Detection + measurement

- compare several classical detectors,
- reject features below a local height threshold,
- compare predictions against manually labelled ground truth.

Preprocessing matters

Raw AFM scan

Preprocessed AFM scan

- AFM images contain global tilt, slowly varying background and scan-line offsets.
- I tested several correction steps, including robust flattening and background subtraction.
- The key trade-off is removing background variation *without suppressing the dots themselves*.

Comparing detection methods

- I implemented and compared several image-processing approaches, including:
  - Laplacian of Gaussian (LoG),
  - Otsu thresholding,
  - Hough circles,
  - anisotropic-diffusion-based methods.
- Each detector produces candidate QDs.
- Candidates are then checked against their **local background**; features below approximately 6 Å in height are rejected.

Insert overlay here

Ground truth vs predicted QDs

How I evaluate the pipeline

A predicted dot is matched to a labelled dot if their centres lie within a small spatial tolerance.

$$\text{Precision} = \frac{TP}{TP + FP},$$
$$\text{Recall} = \frac{TP}{TP + FN},$$
$$F_1 = 2\frac{PR}{P + R}.$$

Insert comparison plot/table here

Precision / Recall / F_1
for each method

This makes the choice of preprocessing and detector quantitative rather than visual.

Where this is going

- The main challenge is not simply finding bright peaks: preprocessing, neighbouring structures and local background all affect the result.
- The current pipeline lets me systematically compare algorithms against the same labelled AFM data.
- Next, I want to use these results to develop a stronger detector, potentially combining classical image features with learned segmentation.

AFM scan → reproducible QD detection → quantitative morphology